

FAIRWEATHER GROUND BUOY 46083, USA

WMO: 996220

Lat: 58.300N

Lon: 137.997W

Elev: 0

StdP: 14.70

Time Zone: -9.00 (W09)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	24.1	27.7	13.9	11.1	27.9	16.9	12.9	31.3	44.0	37.9	41.0	37.8	23.7	130	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.6	61.5	58.3	59.9	57.6	59.0	57.0	59.3	60.2	58.4	59.3	57.3	58.3	9.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.0	74.5	59.6	57.7	71.1	58.6	56.8	68.7	57.9	25.9	60.4	25.2	59.4	24.6	58.6	64.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 40.7	37.5	34.7		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	38.1	38.7	37.4	40.8	45.3	50.7	55.0	56.4	53.4	48.2	42.2	39.6
	DBStd	7.50	5.02	3.69	3.99	2.81	2.96	2.74	2.45	2.25	2.12	2.80	3.99	3.97
	HDD50	2146	367	317	391	277	148	22	0	0	1	66	235	322
	HDD65	7110	832	737	856	727	610	429	311	266	349	521	685	787
	CDD50	511	0	0	0	0	3	43	154	199	102	10	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	15.9	20.8	17.9	17.7	14.7	12.6	11.7	11.0	11.1	14.8	17.3	20.5	20.8
	PrecAvg	101.9	10.4	8.5	7.5	5.4	4.9	3.7	5.2	8.4	12.2	13.4	10.9	11.5
(15) Precipitation	PrecMax	148.1	22.0	15.8	19.7	9.5	9.7	6.9	9.1	15.4	23.0	23.1	18.3	21.9
	PrecMin	80.2	2.3	1.7	1.9	0.9	1.2	1.2	1.9	1.4	3.3	4.3	2.4	6.6
	PrecStd	16.1	4.1	3.7	3.7	2.2	2.4	1.5	1.9	3.5	3.6	4.1	3.8	4.2
	PrecStd	16.1	4.1	3.7	3.7	2.2	2.4	1.5	1.9	3.5	3.6	4.1	3.8	4.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	48.2	47.8	46.1	50.2	54.3	59.1	61.9	64.7	59.7	55.3	50.7	48.4	48.4
	0.4% MCWB	45.1	41.3	41.7	44.6	50.3	56.1	58.0	60.4	55.2	50.7	45.6	43.5	43.5
	2% DB	45.9	45.4	44.1	47.4	52.2	57.3	60.1	62.5	58.4	53.3	49.3	47.3	47.3
	2% MCWB	42.9	41.2	40.5	43.4	48.7	55.2	57.8	59.4	55.1	50.0	45.9	43.4	43.4
	5% DB	44.6	44.2	43.1	45.8	50.4	56.2	59.2	61.0	57.4	52.4	48.2	46.1	46.1
	5% MCWB	41.9	41.2	39.9	42.6	47.2	53.7	57.4	58.2	54.8	49.2	44.8	42.5	42.5
	10% DB	43.6	43.2	42.1	44.4	49.1	54.8	58.4	59.8	56.6	51.6	47.2	44.8	44.8
	10% MCWB	41.5	41.0	39.2	41.5	46.2	52.1	56.5	57.6	54.0	48.6	43.9	41.7	41.7
	0.4% WB	46.2	43.5	43.3	45.8	52.0	56.2	59.5	62.1	57.7	52.9	48.5	46.1	46.1
	0.4% MCDB	47.8	43.7	43.5	48.5	54.1	58.3	59.7	64.9	57.8	53.6	49.5	47.0	47.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	44.1	42.7	41.6	44.2	49.8	54.2	58.8	60.0	56.9	51.8	46.9	44.2	44.2
	2% MCDB	45.1	43.4	42.9	46.8	51.4	55.8	59.2	61.5	57.2	52.6	48.9	45.4	45.4
	5% WB	42.8	42.1	40.7	43.2	47.7	52.6	57.7	59.1	55.7	50.8	45.5	43.0	43.0
	5% MCDB	43.8	43.2	42.3	45.5	49.5	53.7	58.4	60.3	56.4	51.9	47.5	44.5	44.5
	10% WB	42.0	41.3	39.8	42.3	46.7	51.1	56.8	58.1	54.5	49.7	44.4	42.0	42.0
	10% MCDB	43.2	42.9	41.6	44.2	48.6	52.3	57.7	59.4	55.6	51.2	46.7	43.8	43.8
(35) Mean Daily Temperature Range	MDBR	4.6	4.1	4.7	3.7	3.2	2.8	2.6	2.6	3.2	3.7	4.6	4.4	4.4
	5% DB MCDBR	4.7	4.4	4.5	4.4	4.4	3.7	3.5	3.1	3.3	3.9	4.6	4.7	4.7
	5% DB MCWBR	3.7	3.0	3.8	2.8	2.8	2.6	2.8	2.6	3.1	3.4	4.4	4.2	4.2
	5% WB MCDBR	5.0	4.1	4.6	4.0	3.7	2.6	2.9	3.0	3.2	3.3	4.3	5.0	5.0
	5% WB MCWBR	4.6	3.8	4.4	3.0	2.7	2.1	2.7	2.8	3.5	3.5	4.8	5.0	5.0
	5% WB MCWBR	4.6	3.8	4.4	3.0	2.7	2.1	2.7	2.8	3.5	3.5	4.8	5.0	5.0
(40) Clear-Sky Solar Irradiance	taub	0.323	0.306	0.326	0.375	0.389	0.385	0.378	0.376	0.361	0.332	0.301	0.268	0.268
	taud	2.134	2.408	2.384	2.246	2.282	2.356	2.434	2.443	2.434	2.372	2.209	2.143	2.143
	Ebn at Noon	151	235	261	263	267	270	270	262	246	217	161	137	137
	Edh at Noon	17	20	27	36	38	36	32	30	26	21	15	11	11
(44) All-Sky Solar Radiation	RadAvg	114	339	743	1226	1558	1642	1416	1132	762	392	157	74	74
	RadStd	16	49	80	94	141	115	105	147	113	49	21	9	9

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	+2.16	N/A	N/A	N/A	N/A	N/A	-409	+115	N/A

Nomenclature: See separate page